

# PRAKRUT KOTECHA

Robotics Researcher

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prakrutk



## RESEARCH STATEMENT:

My research focuses on developing Physics-Guided World Models that unify analytical mechanics with deep learning to enable truly adaptive robotic control. I aim to formulate architectures, such as Lagrangian Neural Networks (LNNs), that embed fundamental physical symmetries and conservation laws directly into latent dynamics. By integrating these physics-constrained models into hybrid RL-MPC frameworks, I seek to bridge the gap between high-level task planning and low-level inverse dynamics. Ultimately, my goal is to leverage these frameworks for unsupervised skill discovery, allowing robots to autonomously organize diverse motor primitives and complex behaviors without the need for manual reward engineering.

## PUBLICATIONS

### Conference Paper

Prakrut Kotecha, et al. "Neural Predictive Control using Lagrangian Neural Networks for Quadruped Locomotion". 2026 IEEE 22nd Conference on Automation Science and Engineering (CASE)  
✈️ IEEE RAS Travel Grant 2026 [Paper]

Prakrut Kotecha, et al. "Real-Time Gait Adaptation for Quadrupeds using Model Predictive Control and Reinforcement Learning". 2025 Eleventh Indian Control Conference (ICC-11) [Website] [Paper]

Prakrut Kotecha, et al. "Investigating Lagrangian Neural Networks for Infinite Horizon Planning in Quadrupedal Locomotion". 7th International Conference of The Robotics Society, Advances In Robotics 2025  
🏆 Awarded Best Poster Presentation [Paper]

Prakrut Kotecha, et al. "A Hierarchical Manipulation Planning Framework Combining Striking, Pushing, and Pick & Place Motion Primitives". IEEE 20th Conference on Automation Science and Engineering (CASE) 2024  
✈️ IEEE RAS Travel Grant 2024 [Video] [Paper]

### Patent

Prakrut Kotecha, et al. "Navigation System for a vehicle and a method for navigation." published (2019)  
CBR Ref. Number: 201921049473 [Video]

Prakrut Kotecha, et al. "System and Method for Safety-Aware Control of Robot using Predicted Trajectory Evaluation." submitted (2026)  
CBR Ref. Number: 20641116486

## WORKSHOP PAPERS

- Workshop Poster: "PhysFlow: Physics-Grounded Visual World Models via Flow Matching and Lagrangian Neural Networks". at CVPR 2026, Workshop on Embodied AI.
- Workshop Poster: "STRIDE: Structured Lagrangian and Stochastic Residual Dynamics via Flow Matching" at ICRA 2026, Workshop on Contact-Rich Control and Representation

## EDUCATION

### PhD in Cyber-Physical System Indian Institute of Science

📅 August 2023 – Current

Guide: Dr. Shishir Kolathaya  
CGPA: 8.80

Thesis topic: Discovering, Learning, and Deploying Skills through Physics-Guided World Models.

### B.Tech. in Mechanical Engineering Visvesvaraya National Institute of Technology

📅 July 2018 – May 2022

CGPA: 8.07

Bachelor's Thesis Topic: Modelling, Control and Simulation of Quadruped Robot [Thesis]

## OTHER PROJECTS

- Combining RL and MPC for Biped walking [Report] [Video] [Presentation]
- Collision Cone Control Barrier Function for Obstacle Avoidance in UAVs [Presentation]
- Visual Learning Based Optimal Control and Navigation [Report] [Presentation]
- Reconfigurable Robot (QHMuS) - (Snake-Quadruped-Humanoid) [Video]
- Passive Walker [GitHub]
- OsWalT 2.0 (Omni Directional Wall traversing Robot v2.0) [Design Images]

## COURSEWORK

### University Courses:

- E1 277 Reinforcement Learning
- E0 230 Computational Methods of Optimisation
- E1 242 Nonlinear systems and control
- CP 275 Formal Analysis and Control of Autonomous Systems

- Late-breaking Results: "Real-Time Gait Adaptation for Quadrupeds using Model Predictive Control and Reinforcement Learning". at 2025 IEEE-RAS 24th International Conference on Humanoid Robots [\[Website\]](#)

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## EXPERIENCE

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### Loco-Manipulation for Humanoid Robots

#### Advanced Humanoid Research Lab, Infosys Limited

📅 February 2026 - September 2026 📍 Bangalore, India

- Built PointPilot, a training-free, language-conditioned humanoid manipulation system for the Unitree G1 integrating Gemini Robotics-ER 1.6 with G1Pilot/OpenSoT.
- Designed a Control Barrier Function QP safety filter for a torque-sensorless G1 arm, submitted to IEEE Humanoids 2026.
- Contributed to multiple IEEE Humanoids 2026 submissions and supported ongoing patent work in humanoid manipulation.

[\[Website\]](#)

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### Non-Prehensile manipulation

#### International Institute of Information Technology

📅 January 2022 – March 2023 📍 Hyderabad, India

**Advisor:** [Prof. Nagamanikandan G](#) (Assistant Professor at IIIT, Hyderabad)

- Developed a framework for Hybrid Manipulation using a Mobile manipulator.
- Using mathematical program with complementarity constraints did planar pushing for taking a cube from one position to another using only sticking mode of contact.
- Integrated planar pushing and striking with pick-and-place to make a generalised framework.

[\[Video\]](#) [\[Presentation\]](#)

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### Development of control policy for Quadruped Robot

#### Indian Institute of Space Science and Technology

📅 May 2021 – September 2021 📍 Remote Intern

**Advisor:** [Prof. Sam Zachariah](#) (Adjunct Professor at IIST, Scientist at ISRO)

- Worked on a Hybrid dynamics-based controller to produce dynamic walking gait for Quadruped.
- Implemented a controller based on classical control theory using inverse kinematics and dynamics, in Matlab. Further imported the model in Gazebo and tested various gaits in same.

[\[Report\]](#) [\[Presentation\]](#)

### Online Courses:

- MIT 6.832 Underactuated Robotics - Russ Tedrake, MIT
- CS287 Advanced Robotics - Pieter Abbeel, UC Berkeley
- Introduction to Reinforcement Learning - David Silver, DeepMind
- Control of Mobile Robots - Georgia Institute of Technology
- Robotics: Aerial Robotics - University of Pennsylvania

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## EXTRA-CURRICULAR

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- **Presented poster:** at Google Day at IISc 2026.
- **Reviewer:** ICRA, CASE, IROS, ICC
- **Conducted workshop** Walmart Centre For Tech Excellence's Reinforcement Learning Workshop 2025 & 2026 [\[Link\]](#)
- **Attended workshop:** Bangalore Neuromorphic Engineering Workshop [\[Link\]](#)
- **Teaching Assistantship** for Foundation of Robotics Course at IISc (2024 and 2026)
- **Volunteer** for Cyber-Physical Systems Symposium (CyPhySS) 2024.
- **Alumni Secretary** of IvLabs- The Robotics and AI Lab of VNIT.
- **Conducted workshops** on topics such as Image Processing, Microcontrollers, CAD, etc, for 100+ students under the IEEE student chapter.